

Corner singularities obstruct eventual log-concavity of Robin ground states

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Abstract

Let P be a bounded convex polygon that is neither tangential nor a rectangle. We prove that the positive Robin parameters for which the first eigenfunction is log-concave form a locally finite set. In fact, at one fixed obtuse vertex, the eigenfunction and its logarithm fail to be semiconcave on every corner neighborhood for every parameter outside the zero set of a nontrivial real-analytic function. The proof continues a corner singularity of the first variation at the Neumann limit. An explicit angular projection gives an analytic singular coefficient without requiring analytic dependence of a full corner expansion. This disproves the eventual log-concavity conjecture of Andrews, Clutterbuck and Hauer for general bounded convex domains. A single nonrhombic, nonrectangular parallelogram gives a counterexample, and products give examples in every dimension at least two. The result is compatible with eventual concavity on smooth uniformly convex domains.

Keywords: Robin eigenfunction; log-concavity; polygonal domain; corner singularity; analytic perturbation.

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1. Introduction

For a bounded convex polygon $P \subset \mathbb{R}^2$ with nonempty interior, let u_α be the positive first Robin eigenfunction:

$$-\Delta u_\alpha = \lambda_\alpha u_\alpha \quad \text{in } P, \quad \partial_\nu u_\alpha + \alpha u_\alpha = 0 \quad \text{on } \partial P. \quad (1.1)$$

The boundary condition is understood weakly; on the interior of an edge it is classical. We fix the normalization

$$\int_P u_\alpha^2 dx = |P|. \quad (1.2)$$

In particular, $u_0 = 1$ and $\lambda_0 = 0$. Log-concavity means concavity of $\log u_\alpha$ in the open convex set P .

Andrews, Clutterbuck and Hauer [ACH] proved failure of log-concavity for sufficiently small positive Robin parameters on convex polyhedra outside a geometrically distinguished class. Their final Section 10, Conjecture 1, asks whether every fixed bounded convex domain has a threshold above which its Robin ground state is log-concave. Crasta and Fragalà [CF] established eventual log-concavity on

sufficiently smooth uniformly convex domains. Ye and Zhang [YZ] recently reduced the boundary assumption to $C^{3,1}$ in every dimension. These theorems leave open the conjecture on convex domains with corners.

A polygon is *tangential* if it contains a disk tangent to every edge. This is the planar instance of a circumsolid in [ACH]. In dimension two, products of circumsolids are precisely tangential polygons and rectangles. We take polygonal descriptions with no redundant collinear vertices, so every interior angle belongs to $(0, \pi)$.

Theorem 1.1. Let P be a bounded convex polygon that is neither tangential nor a rectangle. There exist an obtuse vertex v_* and a real-analytic function $c : \mathbb{R} \rightarrow \mathbb{R}$ with

$$c(0) = 0, \quad c'(0) \neq 0, \quad (1.3)$$

such that, whenever $\alpha > 0$ and $c(\alpha) \neq 0$, neither u_α nor $\log u_\alpha$ is semiconcave on any sufficiently small neighborhood $P \cap B_r(v_*)$. Consequently

$$\{\alpha > 0 : u_\alpha \text{ is log-concave in } P\} \subseteq \{\alpha > 0 : c(\alpha) = 0\}. \quad (1.4)$$

The set on the left is locally finite, and it is empty for all sufficiently small positive parameters. In particular, for every $A > 0$ there is an $\alpha > A$ for which u_α is not log-concave.

Here semiconcavity on a convex set means that $f(x) - M|x|^2/2$ is concave for some finite constant M . The assertion near v_* refers to each sufficiently small radius, with no uniformity in α required. In fact, the exceptional zero set is finite on every bounded parameter interval, including intervals meeting zero.

Corollary 1.2. The polygon with consecutive vertices

$$(0, 0), \quad (2, 0), \quad (3, 1), \quad (1, 1) \quad (1.5)$$

does not have an eventual log-concavity threshold. There are bounded convex polyhedral counterexamples in every dimension $d \geq 2$.

The new point is the continuation of a regularity obstruction through the entire finite Robin parameter axis. The first-variation classification in [ACH] supplies the initial nonzero corner mode. We extract its continuation by a bounded functional on a fixed Sobolev space, then use the identity theorem for real-analytic functions. Thus the argument does not extrapolate a small-parameter asymptotic estimate to large parameters.

We do not determine whether log-concavity occurs at any of the exceptional parameters. Nor do we prove a classification for tangential polygons. Our result concerns log-concavity and semiconcavity; it does not resolve the higher-dimensional quasiconcavity conjecture or the sharp Robin fundamental-gap problem in [ACH].

2. Analytic dependence and local regularity

We first record precisely the regularity needed for the coefficient construction. Write

$$\Gamma_\omega = \{(r \cos \theta, r \sin \theta) : r > 0, 0 < \theta < \omega\}, \quad S_R = \Gamma_\omega \cap B_R(0), \quad \beta = \frac{\pi}{\omega} > 1. \quad (2.1)$$

Lemma 2.1 (sector regularity). Suppose $z \in H^1(\Gamma_\omega)$ is supported in a fixed closed ball, has homogeneous weak Neumann data on the rays, and $\Delta z \in L^p(\Gamma_\omega)$, where $2 \leq p < \infty$. If

$$2 - \frac{2}{p} < \beta, \quad (2.2)$$

then $z \in W^{2,p}(\Gamma_\omega)$. For a fixed support radius the inclusion is continuous for the graph norm $\|z\|_{H^1} + \|\Delta z\|_{L^p}$.

Proof. This is a direct planar consequence of Dauge's wedge theorem [D, Theorem 8.1], which applies to compactly supported weak Neumann solutions on $\mathbb{R} \times \Gamma_\omega$ when $0 < 2 - 2/p < \pi/\omega$. To check the reduction, choose a nonzero $\eta \in C_c^\infty(\mathbb{R})$ and set $Z(t, x) = \eta(t)z(x)$. Then $Z \in H^1$ has compact support and

$$\Delta_{t,x} Z = \eta \Delta_x z + \eta'' z \in L^p. \quad (2.3)$$

Here $z \in L^p$ follows from the planar H^1 embedding on a bounded sector containing its support. The boundary data of Z are homogeneous Neumann. The cited theorem gives $Z \in W^{2,p}$, hence $z \in W^{2,p}$, for example by integration against a smooth function $\xi(t)$ with $\int \xi \eta = 1$.

For continuity, consider the pairs $(z, f) \in H^1 \times L^p$ with the fixed support restriction and

$$\int_{\Gamma_\omega} \nabla z \cdot \nabla \varphi = - \int_{\Gamma_\omega} f \varphi \quad \text{for all } \varphi \in C_c^\infty(\overline{\Gamma_\omega}). \quad (2.4)$$

They form a closed subspace of $H^1 \times L^p$. The map $(z, f) \mapsto z$ into $W^{2,p}$ has closed graph, since both limits agree as distributions. The closed graph theorem proves the estimate. $\square$

Lemma 2.2 (analytic eigenpair). With normalization (1.2), the maps $\alpha \mapsto \lambda_\alpha$ and $\alpha \mapsto u_\alpha$ are real analytic on $\mathbb{R}$, the latter with values in $H^1(P)$.

Proof. The Robin form

$$a_\alpha(f, g) = \int_P \nabla f \cdot \nabla g + \alpha \int_{\partial P} f g \quad (2.5)$$

has fixed domain $H^1(P)$. The trace map is bounded, and a sufficiently large scalar shift makes the form coercive locally uniformly in α . The first eigenvalue is simple and its eigenfunction can be chosen positive; see [ACH, Proposition 3.1].

For completeness, analytic dependence follows from the analytic implicit function theorem applied to

$$(\alpha, u, \lambda) \mapsto \left(a_\alpha(u, \cdot) - \lambda(u, \cdot)_{L^2(P)}, (u, u)_{L^2(P)} - |P| \right) \quad (2.6)$$

with values in $(H^1(P))^* \times \mathbb{R}$. Its derivative in (u, λ) sends (h, s) to

$$\left(B_\alpha h - s(u_\alpha, \cdot)_{L^2(P)}, 2(u_\alpha, h)_{L^2(P)} \right), \quad B_\alpha = a_\alpha - \lambda_\alpha(\cdot, \cdot)_{L^2(P)}. \quad (2.7)$$

The operator $B_\alpha : H^1 \rightarrow (H^1)^*$ is Fredholm of index zero, by a coercive shift and compactness of the L^2 embedding. Its kernel is the span of u_α , and its range is the annihilator of u_α . Pairing the first component with u_α determines s , the second component determines the component of h along u_α , and B_α is invertible on the complementary subspace. Thus (2.7) is an isomorphism. The local

analytic branches remain the first eigenpair by the spectral gap. Positivity and (1.2) make them agree on overlaps, giving the asserted real-analytic maps. $\square$

At a vertex translated to the origin, let ν_1, ν_2 be the two outward unit normals. They are linearly independent. Let γ be the unique vector satisfying

$$\nu_1 \cdot \gamma = \nu_2 \cdot \gamma = -1. \quad (2.8)$$

Define the local gauge

$$w_\alpha(x) = e^{-\alpha\gamma \cdot x} u_\alpha(x). \quad (2.9)$$

It satisfies homogeneous Neumann conditions on both rays and

$$\Delta w_\alpha = F_\alpha, \quad F_\alpha = -2\alpha\gamma \cdot \nabla w_\alpha - (\lambda_\alpha + \alpha^2|\gamma|^2)w_\alpha. \quad (2.10)$$

Lemma 2.3 (regularity of the gauge). On a sufficiently small fixed sector, w_α depends real analytically on α in H^2 . Consequently F_α is real analytic in L^p for every finite p . For each fixed α , w_α is $C^{1,\delta}$ up to the vertex for some $\delta > 0$, and

$$\nabla w_\alpha(0) = 0. \quad (2.11)$$

Proof. Choose a smooth radial cutoff χ supported strictly inside the radius on which P agrees with the sector, and equal to one on a smaller sector. Extend $z_\alpha = \chi w_\alpha$ by zero to Γ_ω . Its ray normal derivative is zero. Lemma 2.2 and multiplication by the exponential show that z_α is analytic in H^1 . Moreover,

$$\Delta z_\alpha = \chi F_\alpha + 2\nabla \chi \cdot \nabla w_\alpha + (\Delta \chi)w_\alpha \quad (2.12)$$

is analytic in L^2 . The pairs $(z_\alpha, \Delta z_\alpha)$ are therefore analytic in the closed graph space in Lemma 2.1. Since $1 < \beta$, that lemma with $p = 2$ makes z_α analytic in H^2 . Planar Sobolev embedding gives analytic dependence of w_α and ∇w_α in every finite L^p on a smaller sector, and hence the claim about F_α .

Choose $p > 2$ sufficiently close to two that (2.2) holds. A second, nested radial cutoff and (2.12), now with L^p data, give $w_\alpha \in W^{2,p}$ locally. Morrey embedding gives $C^{1,\delta}$ regularity, with $0 < \delta < 1 - 2/p$. The two Neumann conditions then hold at the vertex by continuity. The independent normals force (2.11). $\square$

Lemma 2.4 (positivity at the vertex). If $\alpha > 0$, then $w_\alpha(0) > 0$.

Proof. On a sufficiently small sector put $\rho(x) = e^{2\alpha\gamma \cdot x}$ and $k = \lambda_\alpha + \alpha^2|\gamma|^2 > 0$. Equation (2.10) becomes

$$-\operatorname{div}(\rho \nabla w_\alpha) = k \rho w_\alpha. \quad (2.13)$$

The ground state is positive in the interior. It is also positive at every point in the relative interior of either edge: otherwise the Hopf boundary lemma at that smooth boundary point would contradict the Robin condition. Thus w_α has a positive minimum m on the closed outer circular arc of a small sector. Regularity from Lemma 2.3 gives continuity on its closure.

Use $\varphi = (m - w_\alpha)_+$ as a test function in (2.13), with zero trace on the outer arc and no restriction on the rays. The ray fluxes vanish, and $w_\alpha \geq 0$. Hence

$$-\int_{S_R} \rho |\nabla \varphi|^2 = k \int_{S_R} \rho w_\alpha \varphi \geq 0. \quad (2.14)$$

It follows that $\varphi = 0$, since its outer-arc trace is zero. Therefore $w_\alpha \geq m$ on the whole sector, including its vertex. $\square$

3. A bounded functional for the corner coefficient

Fix an obtuse sector, so $1 < \beta < 2$, and a radius R lying inside the region of Lemma 2.3. Set

$$\begin{aligned} a_\alpha(r) &= \frac{2}{\omega} \int_0^\omega w_\alpha(r, \theta) \cos(\beta\theta) d\theta, \\ b_\alpha(r) &= \frac{2}{\omega} \int_0^\omega F_\alpha(r, \theta) \cos(\beta\theta) d\theta. \end{aligned} \quad (3.1)$$

Lemma 3.1 (analytic coefficient). The limit

$$c(\alpha) = \lim_{r \downarrow 0} r^{-\beta} a_\alpha(r) \quad (3.2)$$

exists for every real α and is a real-analytic function of α . More precisely,

$$\begin{aligned} c(\alpha) &= R^{-\beta} a_\alpha(R) \\ &\quad - \frac{1}{2\beta} \int_0^R (s^{1-\beta} - R^{-2\beta} s^{1+\beta}) b_\alpha(s) ds. \end{aligned} \quad (3.3)$$

Proof. Angular projection of (2.10), using the homogeneous Neumann conditions, gives

$$a_\alpha'' + \frac{1}{r} a_\alpha' - \frac{\beta^2}{r^2} a_\alpha = b_\alpha. \quad (3.4)$$

This equation first holds distributionally away from zero; the functions are smooth there. Choose a finite exponent p so large that

$$p > \frac{2}{2-\beta}, \quad \kappa := 2 - \frac{2}{p} > \beta. \quad (3.5)$$

Lemma 2.3 permits this choice for F_α . Variation of constants in (3.4) gives the regular particular solution

$$A_\alpha(r) = \frac{1}{2\beta} \left[r^\beta \int_0^r s^{1-\beta} b_\alpha(s) ds - r^{-\beta} \int_0^r s^{1+\beta} b_\alpha(s) ds \right]. \quad (3.6)$$

Hölder's inequality, with planar area measure $s ds d\theta$, gives

$$|A_\alpha(r)| + r |A_\alpha'(r)| \leq C r^\kappa \|F_\alpha\|_{L^p(S_r)}. \quad (3.7)$$

For the first integral the relevant weight is $s^{-\beta}$ relative to area measure. Its $L^{p'}$ norm is finite exactly when $\beta p' < 2$, which is equivalent to (3.5). The second integral is less singular. The homogeneous solutions of (3.4) are r^β and $r^{-\beta}$. The latter is excluded by $w_\alpha \in H^1(S_R)$, since its angular projection would have infinite H^1 energy. Thus

$$a_\alpha(r) = c(\alpha)r^\beta + A_\alpha(r). \quad (3.8)$$

Equations (3.7) and (3.5) prove (3.2). Evaluating (3.8) at R proves (3.3).

The trace functional $w \mapsto a(R)$ is bounded on H^2 on a sector extending past R . The integral in (3.3) is a bounded linear functional of $F \in L^p(S_R)$ by the same weighted Hölder estimate. Lemma 2.3 therefore proves analyticity of (3.3). The value is independent of the auxiliary radius because it equals the limit (3.2). $\square$

Two different exponents have been used for different purposes. An exponent just above two yields C^1 regularity in Lemma 2.3. A much larger exponent is allowed for the forcing F_α in Lemma 3.1. No $W^{2,p}$ regularity of w_α is asserted for that larger exponent; generally it is false when $c(\alpha) \neq 0$.

Lemma 3.2 (semiconcavity annihilates the coefficient). For $\alpha > 0$, if either u_α or $\log u_\alpha$ is semiconcave on a neighborhood of the vertex, then $c(\alpha) = 0$.

Proof. By Lemmas 2.3 and 2.4, u_α , w_α and their logarithms are C^1 on a smaller closed sector, and the positive functions are bounded away from zero there. If u_α is semiconcave, its interior Hessian has an upper bound, while

$$\Delta u_\alpha = -\lambda_\alpha u_\alpha \quad (3.9)$$

is bounded. In two dimensions a symmetric matrix with bounded trace and eigenvalues bounded above has eigenvalues bounded below as well. Therefore $D^2 u_\alpha$ is bounded in the interior of that sector. Integration along segments in the convex sector gives $u_\alpha \in C^{1,1}$ up to the vertex, and multiplication by the smooth gauge gives $w_\alpha \in C^{1,1}$.

If $\log u_\alpha$ is semiconcave, put $q = \log w_\alpha = \log u_\alpha - \alpha\gamma \cdot x$. Subtracting a linear function preserves semiconcavity. Equation (2.10) gives

$$\Delta q = -|\nabla q|^2 - 2\alpha\gamma \cdot \nabla q - (\lambda_\alpha + \alpha^2|\gamma|^2). \quad (3.10)$$

Its right hand side is bounded. The same Hessian argument gives $q \in C^{1,1}$ and hence $w_\alpha \in C^{1,1}$.

In either case, (2.11) yields

$$w_\alpha(x) = w_\alpha(0) + O(|x|^2). \quad (3.11)$$

The angular integral of $\cos(\beta\theta)$ is zero, so (3.1) gives $a_\alpha(r) = O(r^2)$. Since $\beta < 2$, (3.2) forces $c(\alpha) = 0$. The same argument works on every smaller corner neighborhood. $\square$

4. The initial nonzero mode

Let

$$v = \left. \frac{du_\alpha}{d\alpha} \right|_{\alpha=0}, \quad \mu = \frac{|\partial P|}{|P|}. \quad (4.1)$$

Differentiation of the weak eigenvalue equation and normalization gives

$$\Delta v = -\mu \quad \text{in } P, \quad \partial_\nu v = -1 \quad \text{on } \partial P, \quad \int_P v = 0. \quad (4.2)$$

We use the following established rigidity result, with regularity on the closure understood: by [ACH, Corollary 8.3], a solution of (4.2) is $C^2(\bar{P})$ if and only if P is a product of circumsolids. Thus it is not $C^2(\bar{P})$ under the hypotheses of Theorem 1.1.

Lemma 4.1 (nonzero first variation). If P is neither tangential nor a rectangle, there is an obtuse vertex for which the coefficient in Lemma 3.1 satisfies $c(0) = 0$ and $c'(0) \neq 0$.

Proof. At any vertex, translate the vertex to zero and choose γ as in (2.8). The function

$$h(x) = v(x) + \frac{\mu}{4}|x|^2 - \gamma \cdot x \quad (4.3)$$

is harmonic with homogeneous Neumann data on the two rays. Indeed, the normal derivative of $|x|^2$ vanishes on either ray. Separation of variables gives

$$h(r, \theta) = d_0 + \sum_{j \geq 1} d_j r^{j\pi/\omega} \cos\left(\frac{j\pi\theta}{\omega}\right). \quad (4.4)$$

This expansion also follows from [ACH, Proposition 5.5]. It can be obtained directly by expanding the trace on an inner arc in the Neumann cosine basis. The H^1 condition excludes negative powers and the logarithmic zero mode. On every strictly smaller sector the series and its derivatives converge away from zero. Moreover, the part with exponents strictly greater than two has second derivatives tending to zero at the vertex: square summability of the trace coefficients and the geometric factor $(r/R)^{j\pi/\omega}$ justify termwise bounds on smaller radii.

If $\omega < \pi/2$, every nonconstant exponent exceeds two. If $\omega = \pi/2$, the first term is a harmonic quadratic polynomial, and all subsequent exponents exceed two. If $\pi/2 < \omega < \pi$, only the first exponent $\beta = \pi/\omega$ lies below two, and the next is $2\beta > 2$. Therefore failure of C^2 regularity at a vertex is possible only at an obtuse vertex with $d_1 \neq 0$.

The solution v is smooth in the interior and up to the relative interiors of edges, by reflection after subtracting a local particular solution with the given normal derivative. If every vertex had a C^2 expansion, these local statements would give $v \in C^2(\bar{P})$, contrary to the rigidity result above. Fix an obtuse vertex with $d_1 \neq 0$.

At this vertex $w_0 = 1$, so $c(0) = 0$. Differentiating (2.9) and (2.10) at zero gives

$$\dot{w}_0 = v - \gamma \cdot x, \quad \dot{F}_0 = -\mu. \quad (4.5)$$

The constant forcing has zero first cosine projection. The radial quadratic term in (4.3) also has zero first cosine projection. It follows from (3.3), which may be differentiated by Lemma 3.1, and (4.4) that

$$c'(0) = d_1 \neq 0. \quad (4.6)$$

This is the claimed vertex and coefficient. $\square$

Proof of Theorem 1.1. Choose the vertex from Lemma 4.1 and its analytic coefficient from Lemma 3.1. Since $c'(0) \neq 0$, this real-analytic function is not identically zero. Its zero set is locally finite on $\mathbb{R}$, by the identity theorem, and zero is an isolated simple zero. Lemma 3.2 shows that semiconcavity of either u_α or its logarithm on any sufficiently small corner neighborhood forces $c(\alpha) = 0$. This proves the local assertion and (1.4). A locally finite subset cannot contain any interval $(A, A + 1)$, so there are arbitrarily large parameters outside the exceptional set. $\square$

5. Explicit domains and the Dirichlet limit

Proof of Corollary 1.2. The polygon (1.5) is a parallelogram with adjacent side lengths 2 and $\sqrt{2}$, and angles $\pi/4$ and $3\pi/4$. It is not a rectangle. A tangential quadrilateral has equal sums of opposite side lengths: the two tangent segments from each vertex have equal lengths, and summing them gives the identity. A tangential parallelogram must therefore have equal adjacent side lengths. Our polygon is not tangential, so Theorem 1.1 applies. At its obtuse vertices the relevant exponent is $\beta = 4/3$.

For $d > 2$, use

$$\Omega_d = P \times (0, 1)^{d-2}. \quad (5.1)$$

The positive product of the first Robin eigenfunctions on the factors satisfies the Robin condition with the same parameter on every face. Its eigenvalue is the sum of the first eigenvalues of the factors, and positivity identifies it with the ground state. Restriction to a slice with fixed interior coordinates in the interval factors is a positive constant multiple of u_α on P . Log-concavity of the product would imply log-concavity on this slice. Thus every parameter excluded for P is excluded for Ω_d . $\square$

The contrast with the Dirichlet limit concerns regularity at the corners. A nonzero coefficient in (3.2) is compatible with convergence in weaker norms and with arbitrarily small amplitude as α increases. For each fixed finite parameter with $c(\alpha) \neq 0$, however, the mode r^β with $\beta < 2$ prevents $C^{1,1}$ regularity of the gauge at the vertex. Semiconcavity would force that regularity through the bounded-trace argument of Lemma 3.2. No uniform C^2 convergence near the polygon's corners is available from a Dirichlet limit alone.

The results of [CF] and [YZ] assume smooth uniformly convex domains. Our polygonal examples satisfy neither boundary hypothesis. Approximating a polygon by smooth domains for a fixed parameter does not produce a single smooth domain with the discrete-exception property proved here, since the approximation can depend on the parameter. The remaining smooth-boundary questions must be treated separately.

The method suggests two further problems. One is to determine the exceptional parameters, or to show their absence for particular nonsymmetric polygons. Another is to understand eventual log-concavity within the tangential class, where the first-variation singular coefficient vanishes and the present initial-mode argument gives no obstruction. These questions are distinct from the conjecture for all convex domains, which Corollary 1.2 answers negatively.

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